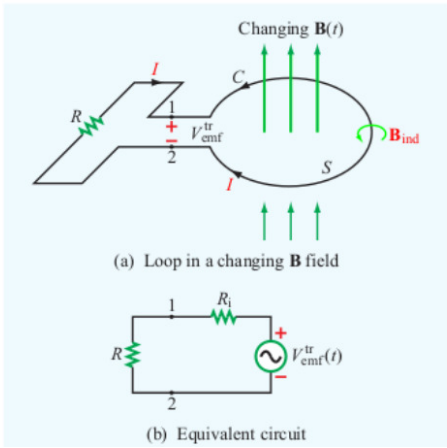
Bao Nguyen ECE 332 Homework #1 Due: 04/21/26

1.) The loop shown below is in the x-y plane and B = z-hat * B0 * sin(ωt).
What is the direction of I (phi-hat or -phi-hat) at the given time?
Explain your answer. (5 points each)



a. t = 0

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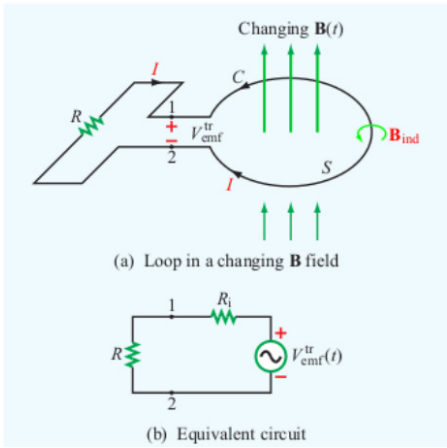

b. ωt = π/4

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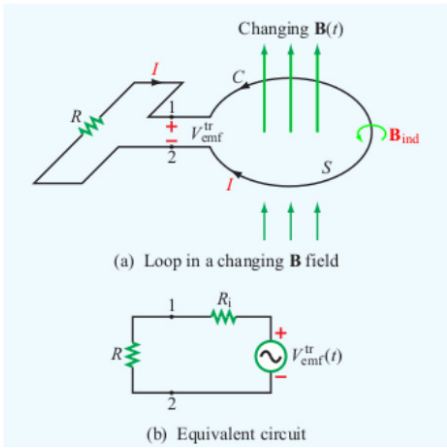
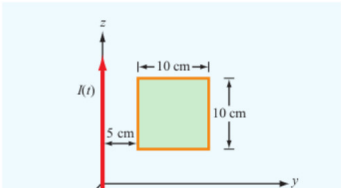
Explain your answer. (5 points each)



c. ωt = π/2

2.) The square loop shown below (10 cm × 10 cm, 5 cm from the wire) is coplanar with a long, straight wire carrying a current:

$I(t) = 5 \cos((2\pi \times 10^4 \text{ rad/s}) t)$ A

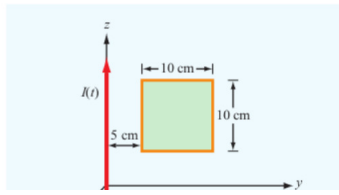


a. (8 points)

Determine the emf induced across a small gap created in the loop.

2.) The square loop shown below (10 cm × 10 cm, 5 cm from the wire) is coplanar with a long, straight wire carrying a current:

$I(t) = 5 \cos((2\pi \times 10^4 \text{ rad/s}) t)$ A



b. (6 points)

Determine the direction and magnitude of the current that would flow through a 4Ω resistor connected across the gap.

The loop has an internal resistance of 1Ω.

3.) The transformer shown below has a long wire along the z-axis carrying I = I0 cos(ωt), coupled to a toroidal coil in the x-y plane centered at the origin. The iron core has relative permeability μr, and 100 turns of tightly wound coil induces voltage V_emf. (15 points total)

(8 points)

(b) Determine the direction and magnitude of the current that flow through a 4Ω resistor connected across the gap. The lc an internal resistance of 1Ω. (6 points)

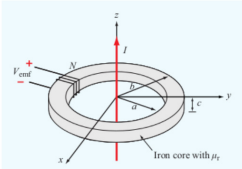


Figure P6.8 Problem 6.8.

a. (10 points)

Develop an expression for V_emf.

3.) The transformer shown below has a long wire along the z-axis carrying $I = I0 \cos(\omega t)$, coupled to a toroidal coil in the x-y plane centered at the origin. The iron core has relative permeability μr, and 100 turns of tightly wound coil induces voltage V_emf. (15 points total)

(8 points)

b) Determine the direction and magnitude of the current that flow through a 4Ω resistor connected across the gap. The lc an internal resistance of 1Ω. (6 points)

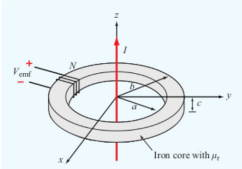


Figure P6.8 Problem 6.8.

2

b. (5 points)

Calculate V_emf for: f = 60 Hz, μr = 4000, a = 5 cm, b = 6 cm, c = 2 cm, I0 = 50 A.

4.) The electromagnetic generator shown below is connected to an electric bulb with resistance 150Ω. The loop area is 0.1 m² and rotates at 3,600 rpm in a uniform magnetic flux density B0 = 0.4 T.
Determine the amplitude of the current generated in the light bulb. (6 points)

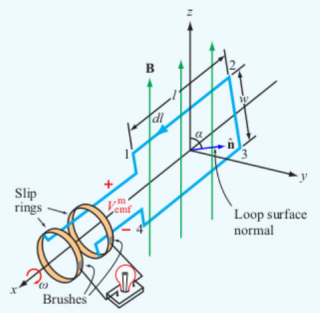


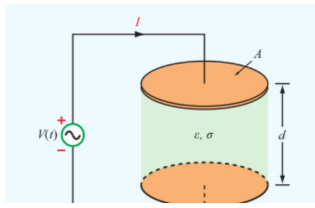
Figure 6-12 A loop rotating in a magnetic field induces an emf.

5.) The plates of a parallel-plate capacitor have areas of 10 cm^2 each and are separated by 2 cm. The capacitor is filled with a dielectric material with $\varepsilon = 4\varepsilon_0$, and the voltage across it is:

$$V(t) = 30 \cos((2\pi \times 10^6 \text{ rad/s}) t) \text{ V}$$

Find the displacement current. (6 points)

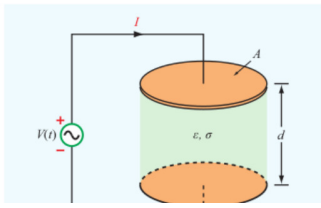
6.) The parallel-plate capacitor shown below is filled with a lossy dielectric of relative permittivity εr and conductivity σ. Plate separation is d, each plate has area A, connected to time-varying voltage V(t). (24 points total)



a. (6 points)

Obtain an expression for Ic, the conduction current flowing between the plates, in terms of the given quantities.

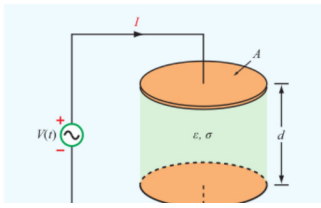
6.) The parallel-plate capacitor shown below is filled with a lossy dielectric of relative permittivity εr and conductivity σ. Plate separation is d, each plate has area A, connected to time-varying voltage V(t). (24 points total)



b. (6 points)

Obtain an expression for Id, the displacement current flowing inside the capacitor.

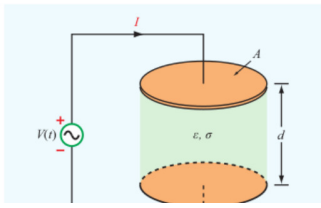
6.) The parallel-plate capacitor shown below is filled with a lossy dielectric of relative permittivity εr and conductivity σ. Plate separation is d, each plate has area A, connected to time-varying voltage V(t). (24 points total)



c. (6 points)

Based on (a) and (b), give an equivalent-circuit representation for the capacitor. Draw and label the circuit.

6.) The parallel-plate capacitor shown below is filled with a lossy dielectric of relative permittivity εr and conductivity σ. Plate separation is d, each plate has area A, connected to time-varying voltage V(t). (24 points total)



d. (6 points)

Evaluate the circuit element values for:

A = 4 cm², d = 0.5 cm, εr = 4, σ = 2.5 S/m,

V(t) = 10 cos((3π × 10^3 rad/s) t) V

7.) At $t = 0$, charge density $\rho v0$ was introduced into the interior of a material with relative permittivity $\varepsilon r = 9$.
If at $t = 1\ \mu s$ the charge density has dissipated to $10^{-3} * \rho v0$, what is the conductivity of the material? (6 points)

8.) If we characterize how good a material is as an insulator by its resistance to dissipating charge, which of the following is the better insulator? (6 points)

Material	**ε_r**	**σ (S/m)**
Dry Soil	2.5	10^-4
Fresh Water	80	10^-3

9.) The electric field of an electromagnetic wave propagating in air is:

E(z,t) = [4 cos((6×10^8 rad/s)t – (2 m^-1)z) x-hat
+ 3 sin((6×10^8 rad/s)t – (2 m^-1)z) y-hat] V/m

Find the associated magnetic field H(z, t). (6 points)

10.) Given the electric field:

$E = E0 \sin(ay) \cos(\omega t - kz)$ x-hat

where E0, a, ω, and k are constants, find H. (8 points)